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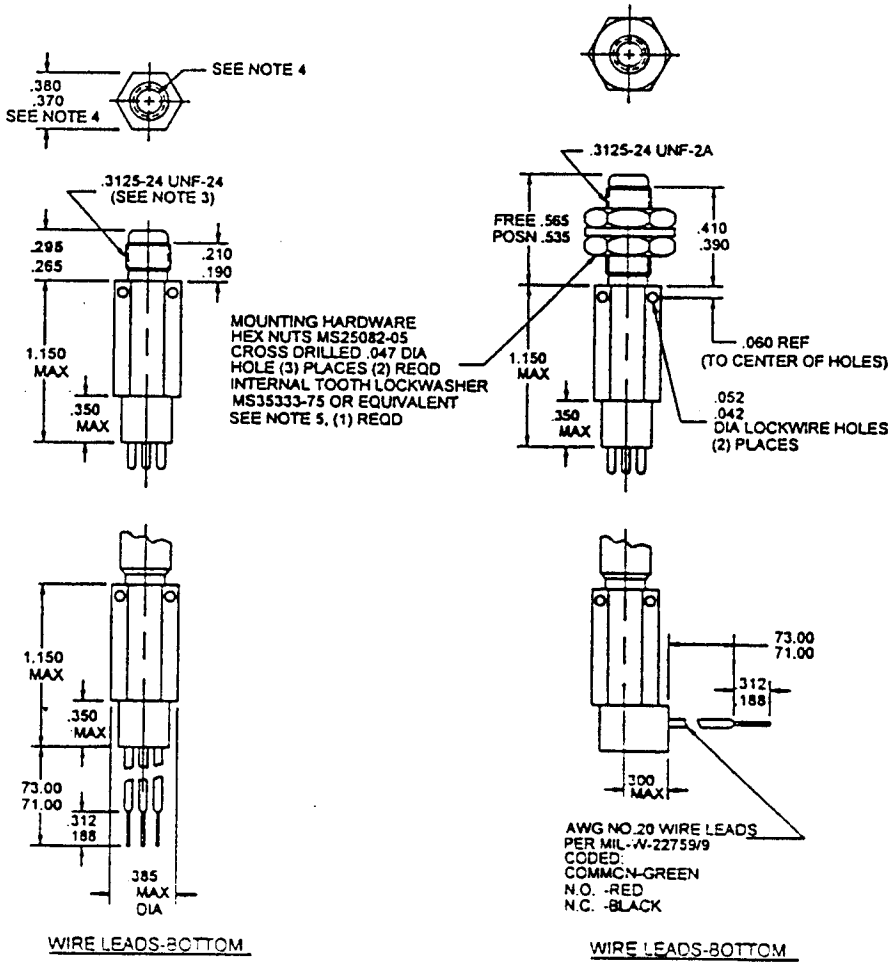
INCH-POUND
MIL-PRF-8805/72H
12 May 1995
SUPERSEDING
MIL-S-8805/72G
7 February 1992

PERFORMANCE SPECIFICATION SHEET

SWITCHES, SENSITIVE, SPDT, PLUNGER, 7 AMPERES AND 4 AMPERES, HERMETICALLY SEALED CONTACTS, UNSEALED PLUNGER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.

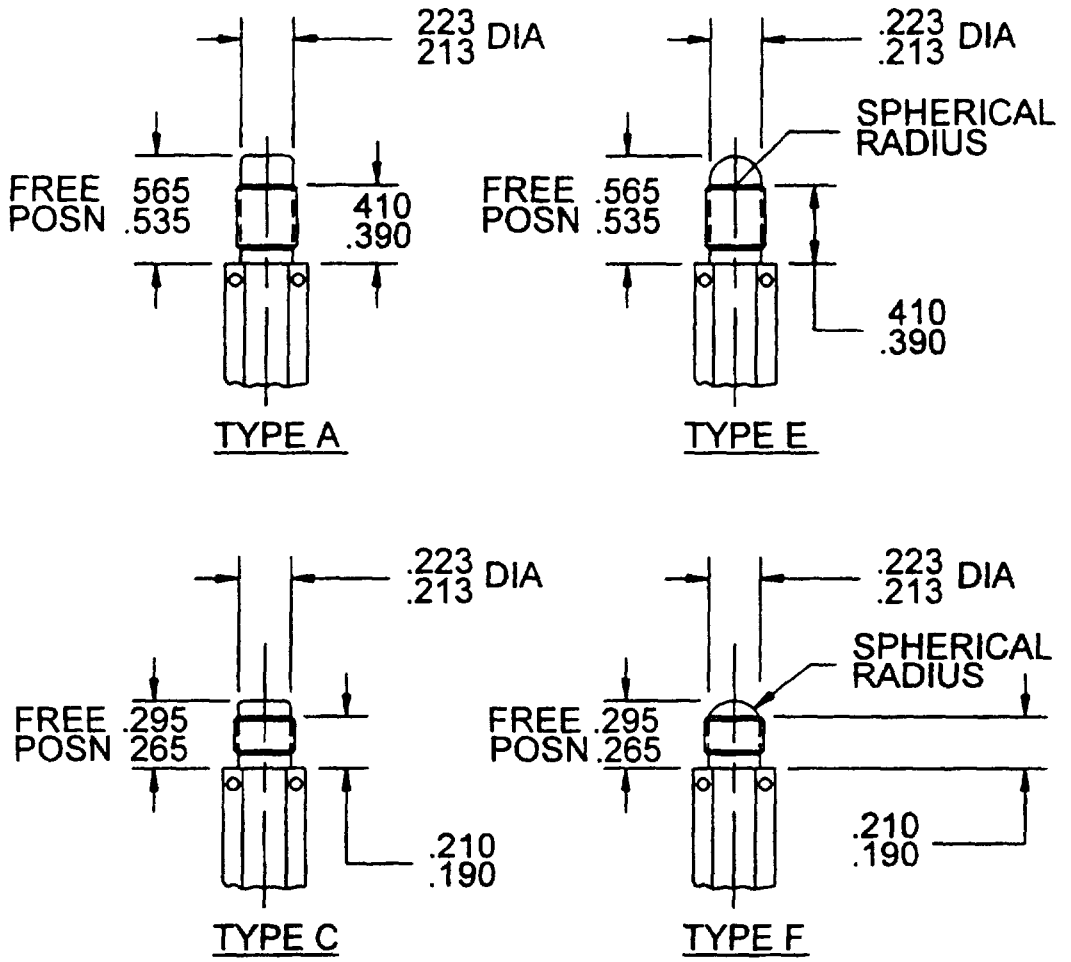


Ⓗ FIGURE 1. Dimensions and configurations.

Ⓗ denotes changes

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inches	mm	inches	mm	inches	mm	inches	mm
.042	1.07	.213	5.41	.350	8.80	.565	14.35
.047	1.19	.223	5.66	.370	9.40	1.150	29.21
.052	1.32	.265	6.73	.380	9.65	71.00	1803.4
.060	1.52	.295	7.49	.385	9.78	73.00	1854.2
.188	4.78	.300	7.62	.390	9.91		
.190	4.83	.312	7.92	.410	10.41		
.210	5.33	.3125	7.94	.535	13.59		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Mounting hardware not supplied on types C and F
4. Body dimension on switches is .375 ±.005 (9.53 ±.13 mm) across the flats and the button diameter is .218 ±.005 (5.54 ±.13 mm).
(H) 5. Alternative base metals and protective finishes, as approved by the qualifying activity, may be utilized for lock washer material. Dimensions shall be inaccordance with the referenced hardware specifications.

FIGURE 1 Dimensions and configurations - Continued

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

(H) Enclosure design: 5 (hermetic). Basic switch shall be hermetically sealed, M8805/68-008 or -022 listed on QPL-8805. Plunger is unsealed.

Temperature characteristic: 2 (-65°C to +125°C)

Shock type: M (100 G, method 213, test condition I, MIL-STD-202).

Vibration grade: 3 (10 - 3,000 Hz swept sinusoidal).

Operating characteristics:

Actuating force: 30 ounces maximum at standard temperature and pressure (21°C and 14.7 psia).

Releasing force: 4 ounces minimum at standard temperature and pressure (21°C and 14.7 psia).

Movement differential: .005 inch (0.13 mm) maximum.

Pretravel: .010 inch (0.25 mm) maximum.

Overtravel:

(H) Types A and E: .125 inch (3.18 mm) minimum.

(H) Types C and F: .060 inch (1.52 mm) minimum.

Contact resistance: .040 ohm plus 0.011 ohm per lead per foot.

Dielectric withstanding voltage:

Sea level and 70,000 feet: 1,050 V rms (terminal to case), 800 V rms (terminal to terminal).

Seal test: Not required.

Mechanical endurance: 50,000 cycles.

Electrical endurance: 25,000 cycles.

Intermediate current: 25,000 cycles.

Low level circuit: 25,000 cycles.

Electrical ratings: See table I.

Short circuit: 20 times rated resistive load (MS3320-7 1/2 used for circuit protection).

Weight: 0.6 ounce + 0.1 ounce per lead per foot.

Qualification:

Group submission: See table II.

Military Part or Identifying Number (PIN): See tables III and IV.

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(M) TABLE I Electrical ratings

Load	Sea level and 70,000 feet 28 V dc (Amperes)	
Resistive	7	4
Inductive	1	1
Lamp	2	2

(M) TABLE II. Qualification.

Inspection	Basic Switch	Additional inspection	Extent of Approval
Qualification inspection table of MIL-S-8805, groups I through VI inclusive (seal test not required)	M8805/72-021, 029 (all sample units) (021 will qualify the 029 because of amp rating but not the reverse)	M8805/72-017 through 020, 022 through 028 030, 031, 032 (2 sample units each) Visual and mechanical inspection only	All

(M) TABLE III. Part numbers and configuration.

PIN 1/	Load Rating	Type of actuating mechanism	Terminals
M8805/72-017	7 Amps	A	Wire leads-bottom
M8805/72-018	7 Amps	C	Wire leads-bottom
M8805/72-019	7 Amps	A	Wire leads-side
M8805/72-020	7 Amps	C	Wire leads-side
M8805/72-021	7 Amps	E	Wire leads-bottom
M8805/72-022	7 Amps	F	Wire leads-bottom
M8805/72-023	7 Amps	E	Wire leads-side
M8805/72-024	7 Amps	F	Wire leads-side
M8805/72-025	4 Amps	A	Wire leads-bottom
M8805/72-026	4 Amps	C	Wire leads-bottom
M8805/72-027	4 Amps	A	Wire leads-side
M8805/72-028	4 Amps	C	Wire leads-side
M8805/72-029	4 Amps	E	Wire leads-bottom
M8805/72-030	4 Amps	F	Wire leads-bottom
M8805/72-031	4 Amps	E	Wire leads-side
M8805/72-032	4 Amps	F	Wire leads-side

1/ Part numbers M8805/72-001 through -016 have been canceled.

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(H) TABLE IV. Substitution information for canceled PIN's. 1/

Canceled PIN's M8805/72-	Substitute PIN's (7 Amp) M8805/72-	Substitute PIN's (4 Amp) M8805/72-
005	017	025
007	018	026
009	019	027
011	020	028
013	021	029
014	022	030
015	023	031
016	024	032

1/ The canceled PIN's are no longer acquirable.The substitute parts are interchangeable in all respects except electrical ratings. The canceled parts were rated at sea level and 70,000 feet altitude as follows:

28 V dc: 7 A resistive, 3 A inductive, 2 A lamp load.
115 V ac, 400 Hz: 5 A resistive, 3 A inductive, 2 A lamp load.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AV, MI
Navy - AS, OS, MC, SH
Air Force - 19, 99

Preparing activity:
DLA - ES

(Project 5930-1594-03)

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